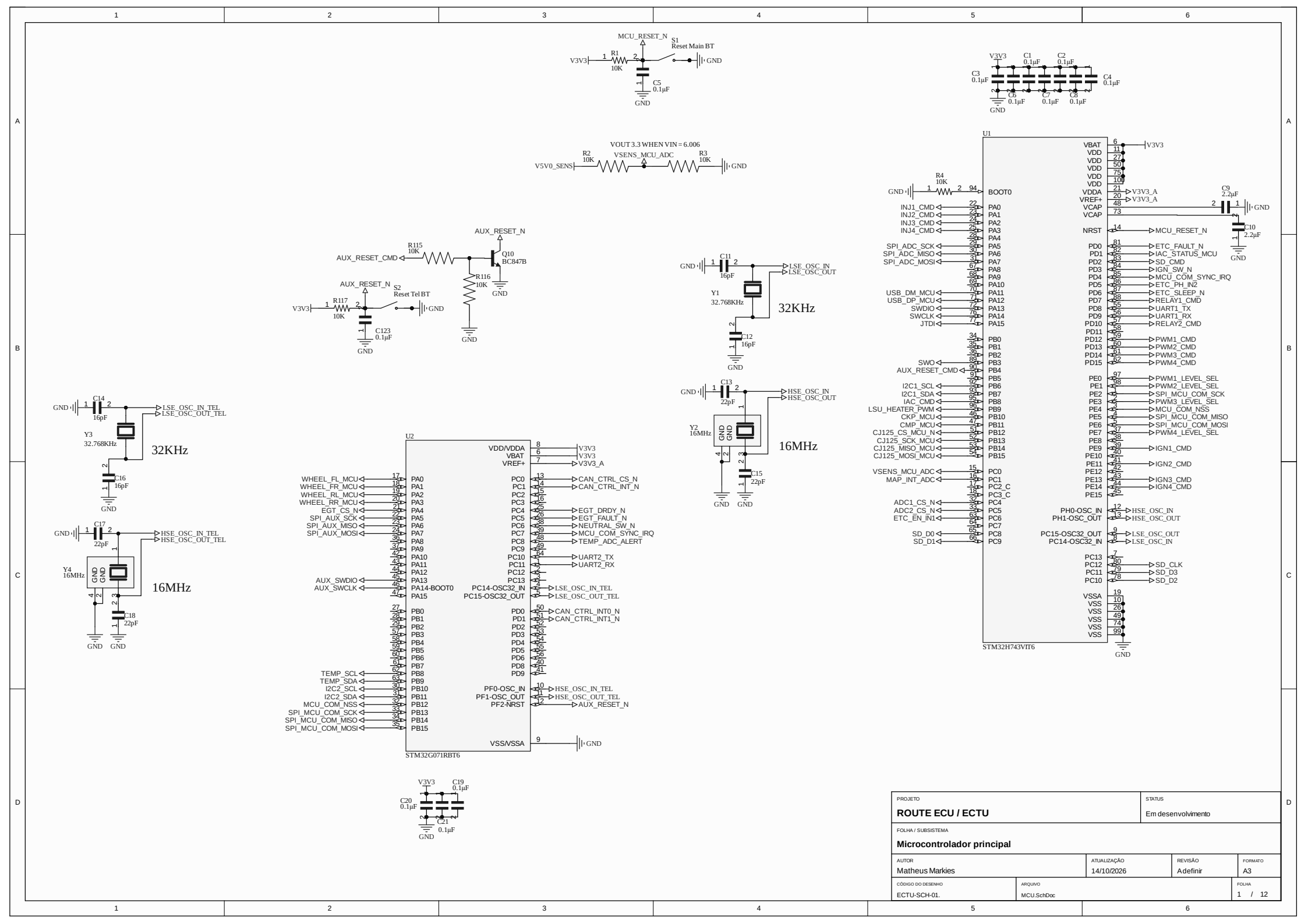
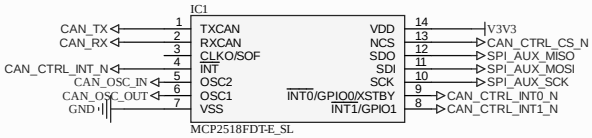
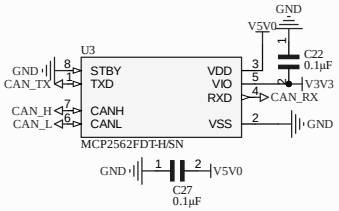
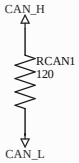


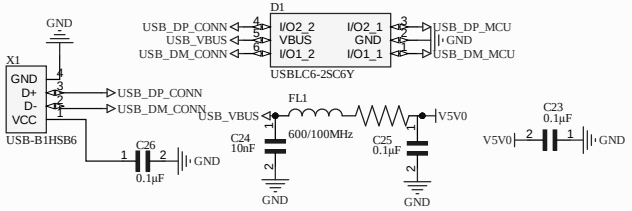
PROJETO	**ROUTE ECU / ECTU**
STATUS	Em desenvolvimento
FOLHA / SUBSISTEMA	**Microcontrolador principal**
AUTOR	Matheus Markies
ATUALIZAÇÃO	14/10/2026
REVISÃO	A definir
FORMATO	A3
CÓDIGO DO DESENHO	ECTU-SCH-01.
ARQUIVO	MCU.SchDoc
FOLHA	1 / 12

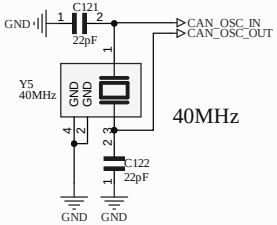
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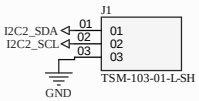


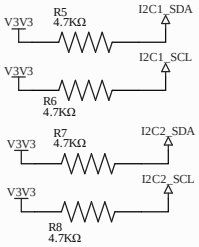




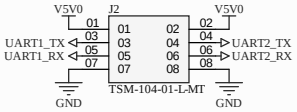
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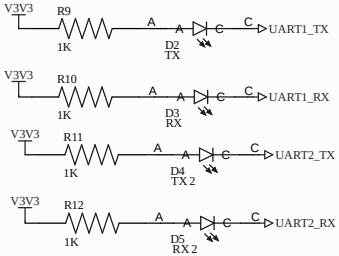
I2C

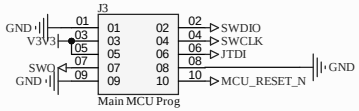


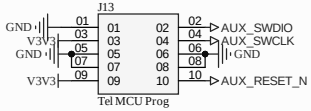


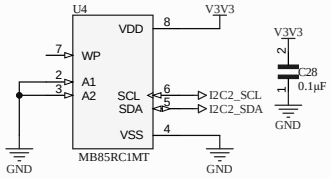
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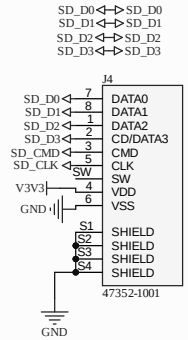




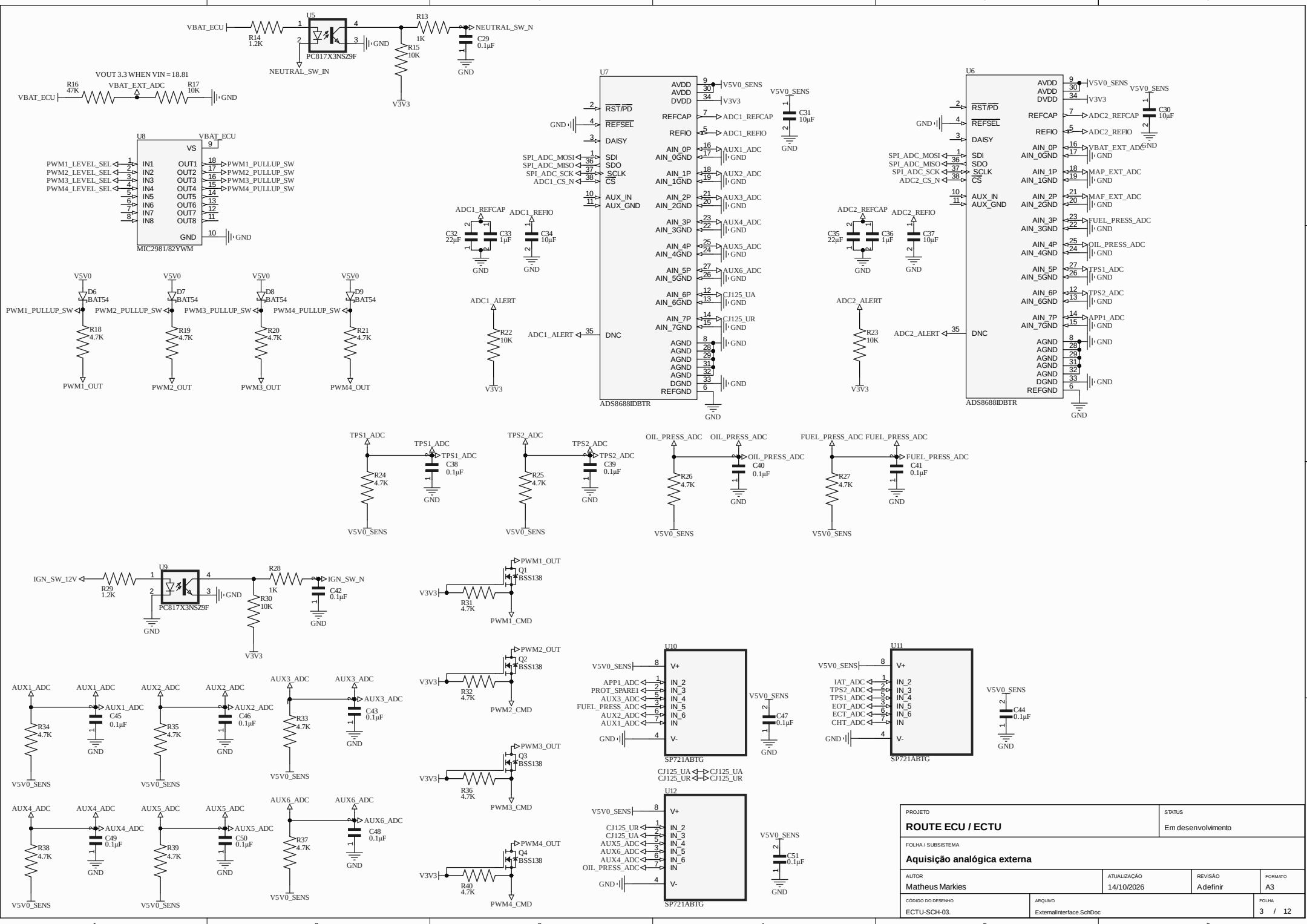


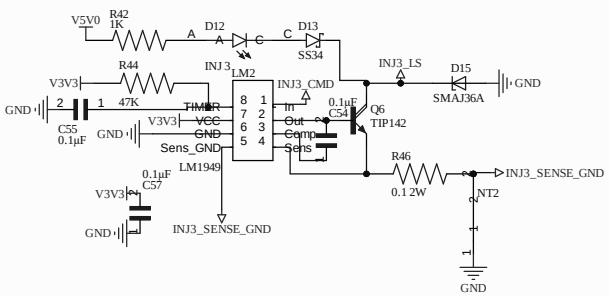


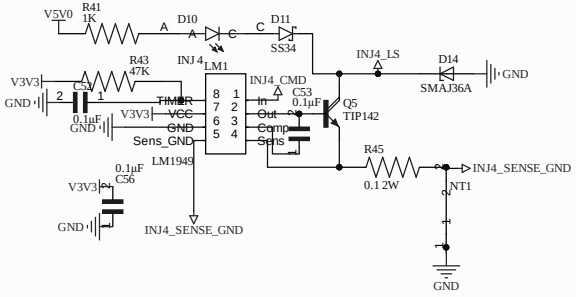


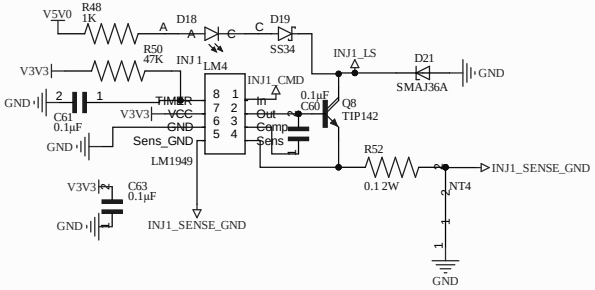


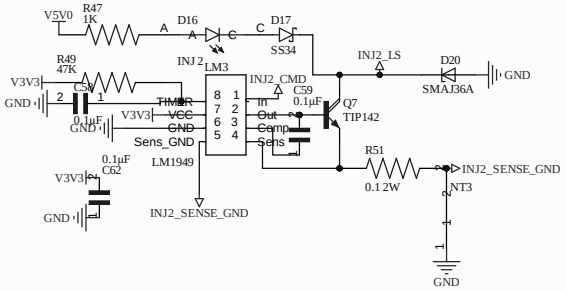
PROJETO: **ROUTE ECU / ECTU**		STATUS: Em desenvolvimento	
FOLHA / SUBSISTEMA: **Interfaces de comunicação**			
AUTOR: Matheus Markies	ATUALIZAÇÃO: 14/10/2026	REVISÃO: A definir	FORMATO: A3
CÓDIGO DO DESENHO: ECTU-SCH-02.	ARQUIVO: CommunicationInterfaces.SchDoc		FOLHA: 2 / 12

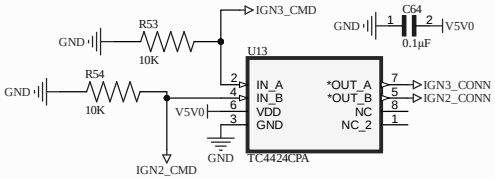

VBAT_ECU
R14 1.2K
U5
PC817X3NSZ9F
NEUTRAL_SW_IN
R13 1K
R15 10K
C29 0.1µF
NEUTRAL_SW_N
GND
V3V3
VOUT 3.3 WHEN VIN = 18.81
R16 47K
VBAT_EXT_ADC
R17 10K
U8
MIC2981/82YWM
PWM1_LEVEL_SEL
PWM2_LEVEL_SEL
PWM3_LEVEL_SEL
PWM4_LEVEL_SEL
PWM1_PULLUP_SW
PWM2_PULLUP_SW
PWM3_PULLUP_SW
PWM4_PULLUP_SW
U7
ADS8688IDBTR
U6
ADS8688IDBTR
V5V0_SENS
V3V3
ADC1_REFCAP
ADC1_REFIO
ADC2_REFCAP
ADC2_REFIO
SPI_ADC_MOSI
SPI_ADC_MISO
SPI_ADC_SCK
ADC1_CS_N
ADC2_CS_N
ADC1_ALERT
ADC2_ALERT
D6 BAT54
D7 BAT54
D8 BAT54
D9 BAT54
R18 4.7K
R19 4.7K
R20 4.7K
R21 4.7K
R22 10K
R23 10K
PWM1_OUT
PWM2_OUT
PWM3_OUT
PWM4_OUT
TPS1_ADC
TPS2_ADC
OIL_PRESS_ADC
FUEL_PRESS_ADC
R24 4.7K
R25 4.7K
R26 4.7K
R27 4.7K
C38 0.1µF
C39 0.1µF
C40 0.1µF
C41 0.1µF
IGN_SW_12V
U9
PC817X3NSZ9F
R29 1.2K
R28 1K
R30 10K
C42 0.1µF
IGN_SW_N
Q1 BSS138
Q2 BSS138
Q3 BSS138
Q4 BSS138
R31 4.7K
R32 4.7K
R36 4.7K
R40 4.7K
PWM1_CMD
PWM2_CMD
PWM3_CMD
PWM4_CMD
AUX1_ADC
AUX2_ADC
AUX3_ADC
AUX4_ADC
AUX5_ADC
AUX6_ADC
R34 4.7K
R35 4.7K
R33 4.7K
R38 4.7K
R39 4.7K
R37 4.7K
C45 0.1µF
C46 0.1µF
C43 0.1µF
C49 0.1µF
C50 0.1µF
C48 0.1µF
U10
U11
U12
SP721ABTG
CJ125_UA
CJ125_UR
APP1_ADC
PROT_SPARE1
FUEL_PRESS_ADC
IAT_ADC
EOT_ADC
ECT_ADC
CHT_ADC
C47 0.1µF
C44 0.1µF
C51 0.1µF
PROJETO
ROUTE ECU / ECTU
STATUS
Em desenvolvimento
FOLHA / SUBSISTEMA
Aquisição analógica externa
AUTOR
Matheus Markies
ATUALIZAÇÃO
14/10/2026
REVISÃO
Adefinir
FORMATO
A3
CÓDIGO DO DESENHO
ECTU-SCH-03.
ARQUIVO
ExternalInterface.SchDoc
FOLHA
3 / 12

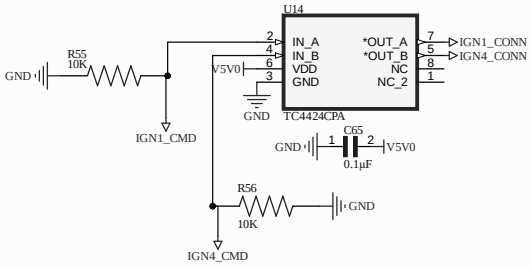




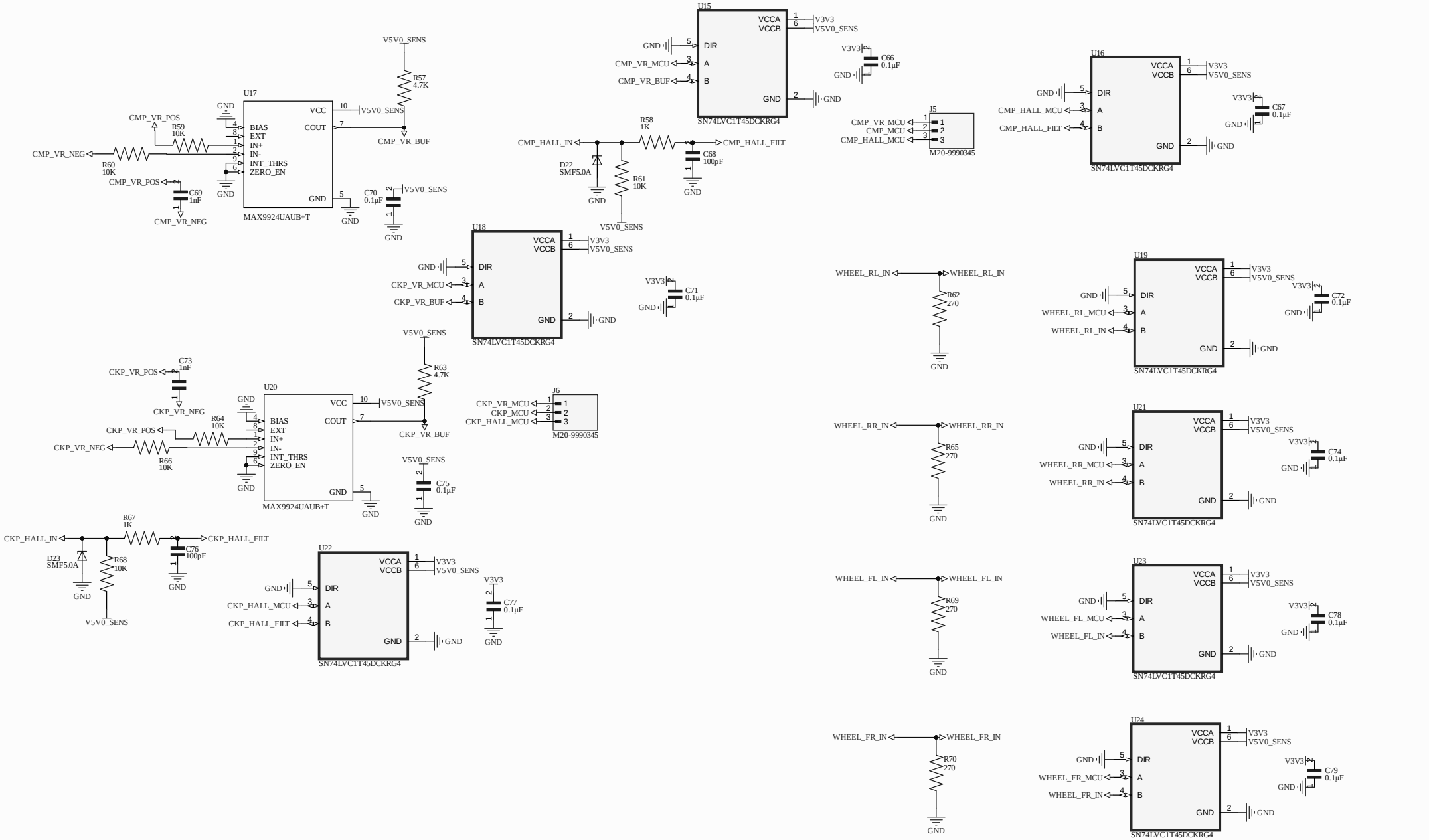






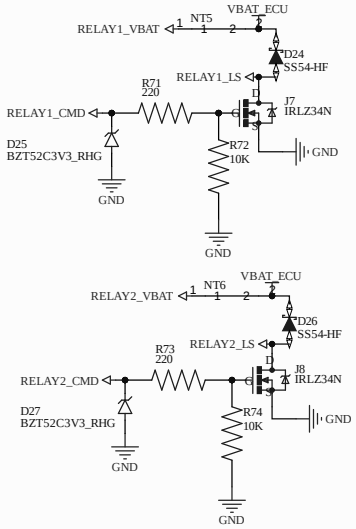


PROJETO: **ROUTE ECU / ECTU**			STATUS: Em desenvolvimento
FOLHA / SUBSISTEMA: **Injeção e ignição**			
AUTOR: Matheus Markies	ATUALIZAÇÃO: 14/10/2026	REVISÃO: A definir	FORMATO: A3
CÓDIGO DO DESENHO: ECTU-SCH-04.	ARQUIVO: SparkAndFuel.SchDoc		FOLHA: 4 / 12

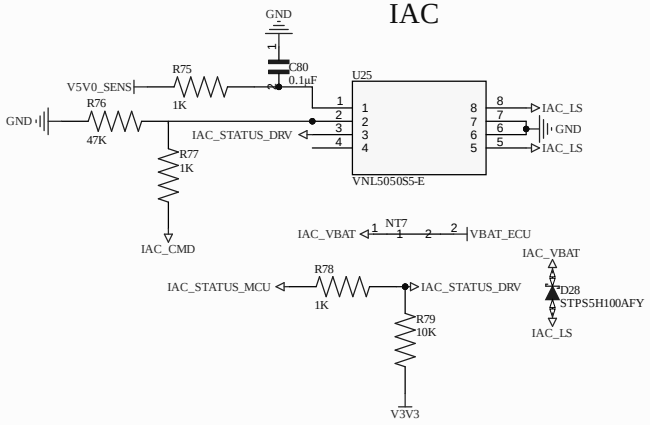


PROJETO	**ROUTE ECU / ECTU**
STATUS	Em desenvolvimento
FOLHA / SUBSISTEMA	**Rotação e fase (CKP / CMP)**
AUTOR	Matheus Markies
ATUALIZAÇÃO	14/10/2026
REVISÃO	A definir
FORMATO	A3
CÓDIGO DO DESENHO	ECTU-SCH-05.
ARQUIVO	CrankAndCam.SchDoc
FOLHA	5 / 12

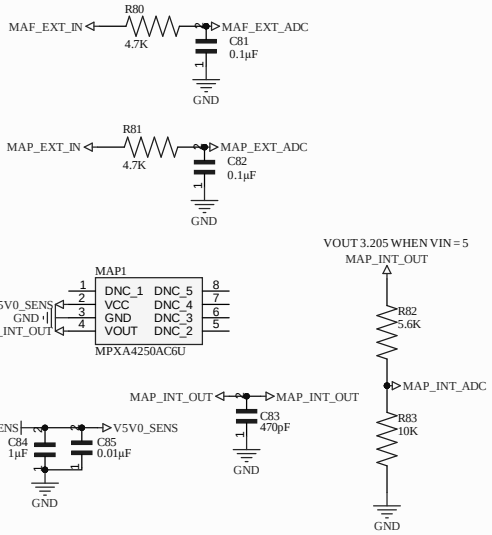
Fuel And Starter



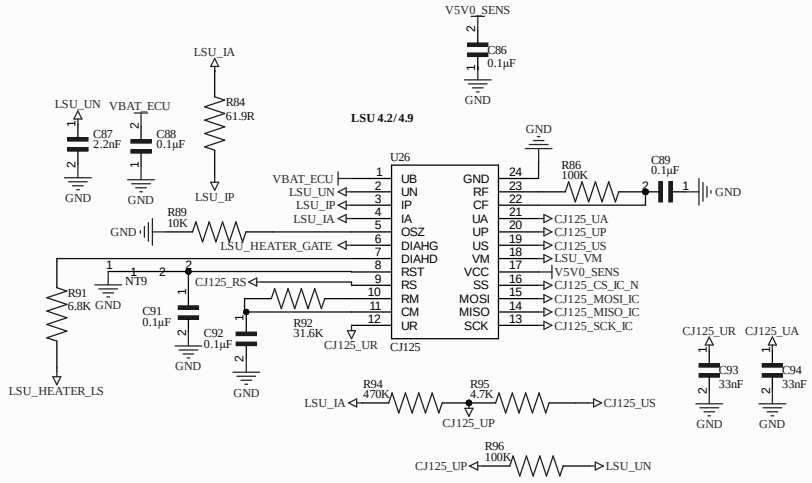
IAC

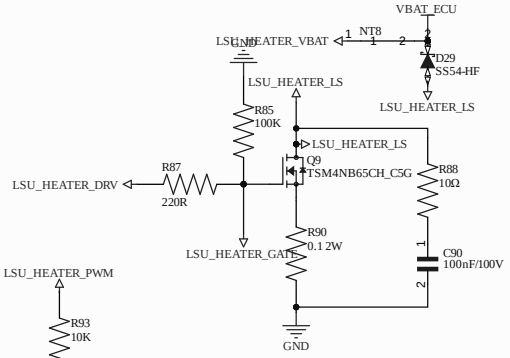


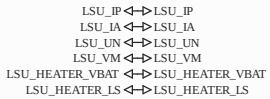
PROJETO: **ROUTE ECU / ECTU**	STATUS: Em desenvolvimento
FOLHA / SUBSISTEMA: **Acionamentos auxiliares**	
AUTOR: Matheus Markies	ATUALIZAÇÃO: 14/10/2026
REVISÃO: A definir	FORMATO: A3
CÓDIGO DO DESENHO: ECTU-SCH-06.	ARQUIVO: LowSideSwitches.SchDoc
FOLHA: 6 / 12	

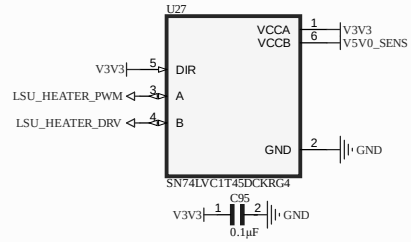


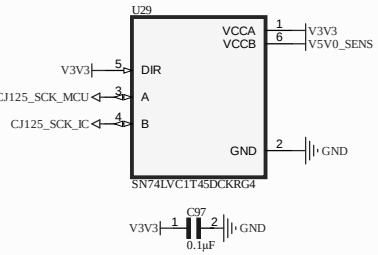
PROJETO: **ROUTE ECU / ECTU**			STATUS: Em desenvolvimento
FOLHA / SUBSISTEMA: **Pressão e fluxo de ar**			
AUTOR: Matheus Markies	ATUALIZAÇÃO: 14/10/2026	REVISÃO: A definir	FORMATO: A3
CÓDIGO DO DESENHO: ECTU-SCH-07.	ARQUIVO: PressureAirflow.SchDoc		FOLHA: 7 / 12

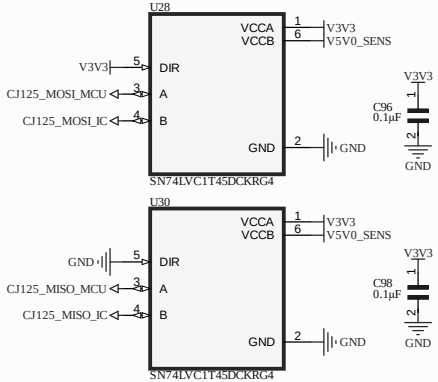


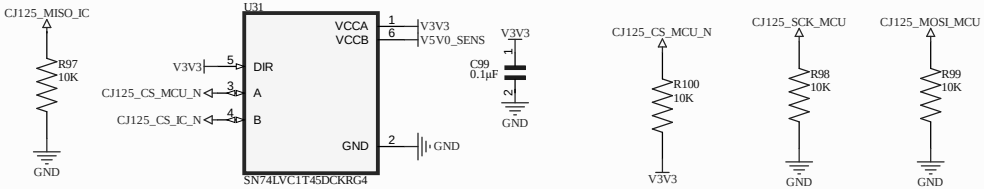




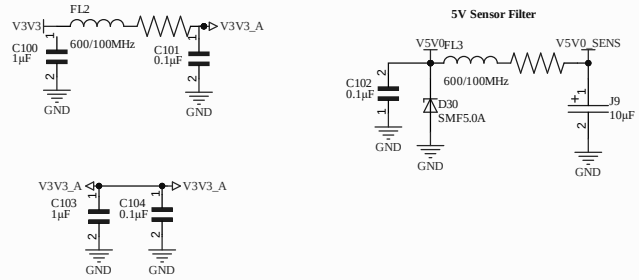


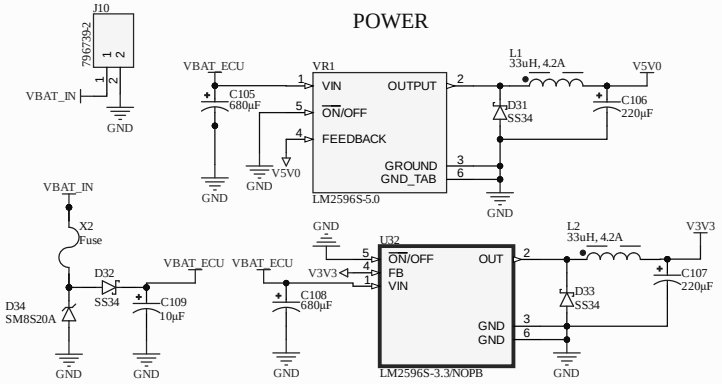


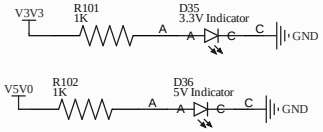




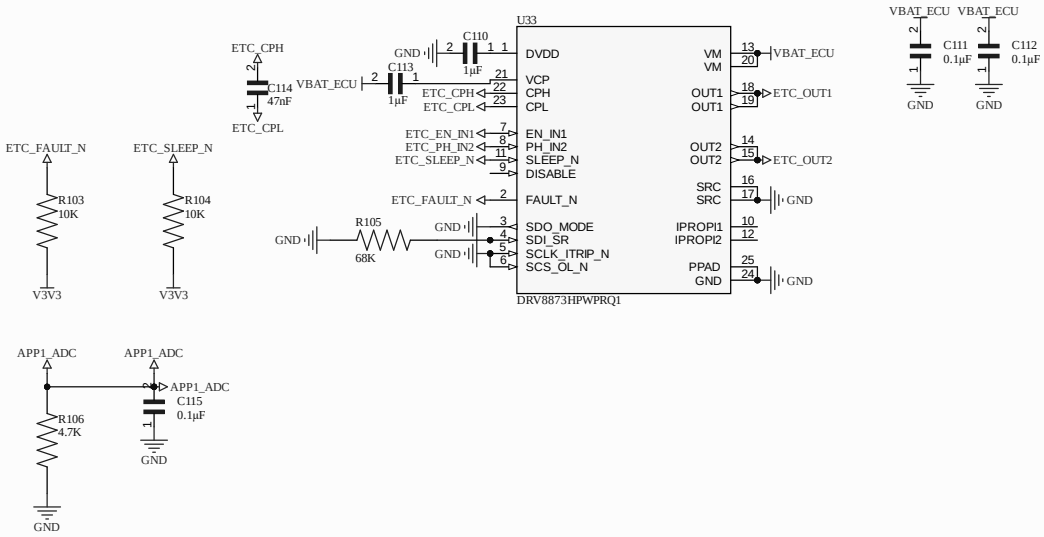
PROJETO: **ROUTE ECU / ECTU**			STATUS: Em desenvolvimento
FOLHA / SUBSISTEMA: **Controle da sonda lambda**			
AUTOR: Matheus Markies	ATUALIZAÇÃO: 14/10/2026	REVISÃO: Adefinir	FORMATO: A3
CÓDIGO DO DESENHO: ECTU-SCH-08.	ARQUIVO: LambdaWidebandController.SchDoc		FOLHA: 8 / 12







PROJETO: **ROUTE ECU / ECTU**			STATUS: Em desenvolvimento
FOLHA / SUBSISTEMA: **Alimentação e proteções**			
AUTOR: Matheus Markies	ATUALIZAÇÃO: 14/10/2026	REVISÃO: Adefinir	FORMATO: A3
CÓDIGO DO DESENHO: ECTU-SCH-09.	ARQUIVO: PowerManagement.SchDoc		FOLHA: 9 / 12

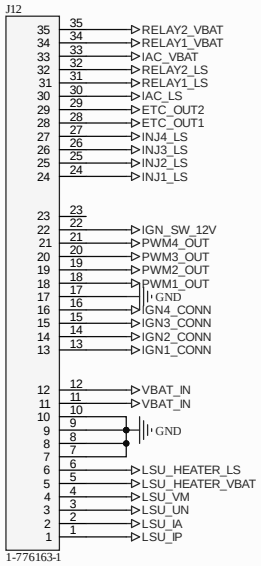


PROJETO	STATUS
ROUTE ECU / ECTU	Em desenvolvimento
FOLHA / SUBSISTEMA	
Controle eletrônico da borboleta	

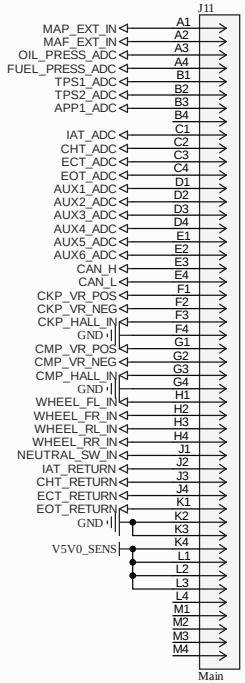
AUTOR	ATUALIZAÇÃO	REVISÃO	FORMATO
Matheus Markies	14/10/2026	A definir	A3

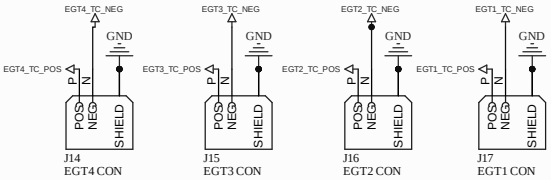
CÓDIGO DO DESENHO	ARQUIVO	FOLHA
ECTU-SCH-10.	ETC.SchDoc	10 / 12

J12 | ATUACAO / ALIMENTACAO

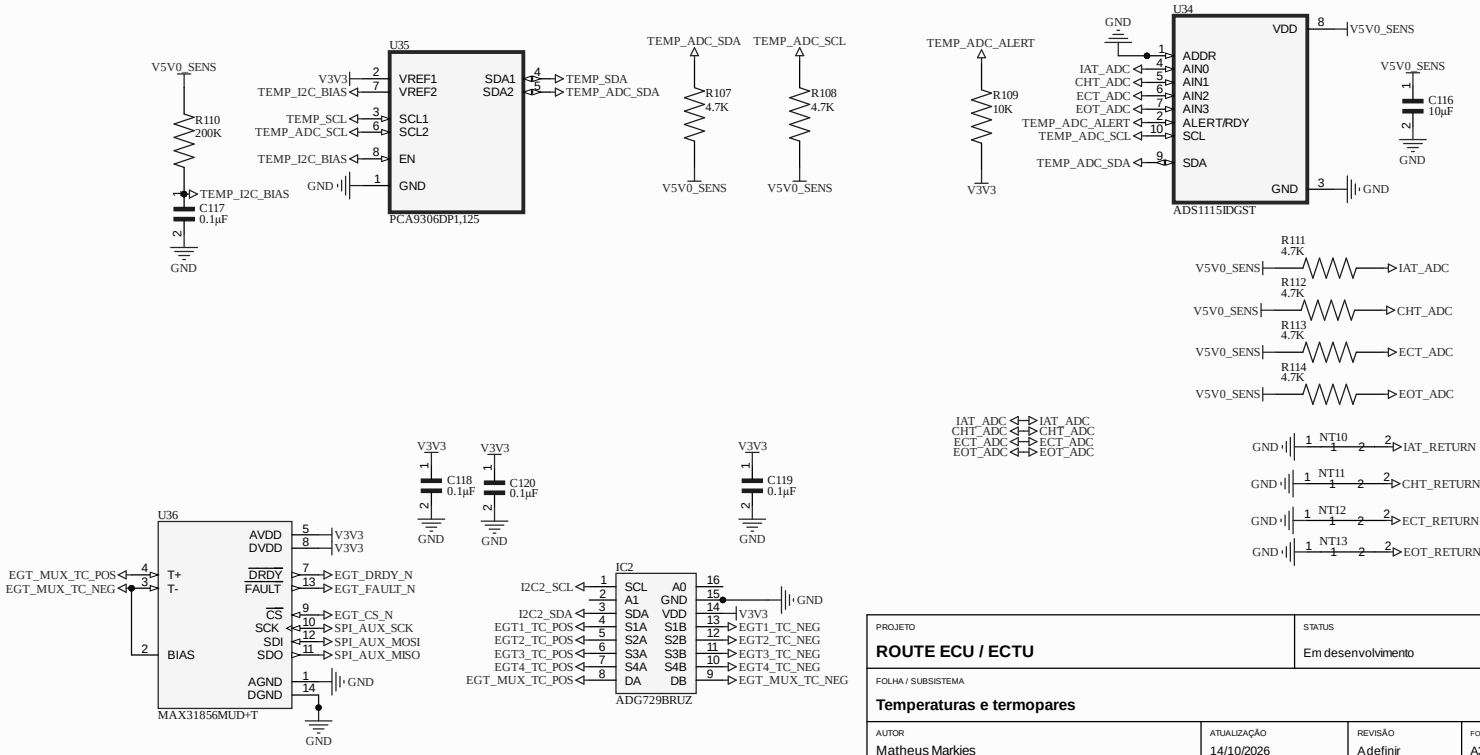


J11 | SENSORES / COMUNICACAO





PROJETO: **ROUTE ECU / ECTU**			STATUS: Em desenvolvimento
FOLHA / SUBSISTEMA: **Conectores da ECU**			
AUTOR: Matheus Markies	ATUALIZAÇÃO: 14/10/2026	REVISÃO: A definir	FORMATO: A3
CÓDIGO DO DESENHO: ECTU-SCH-11.	ARQUIVO: ECUConnectors.SchDoc		FOLHA: 11 / 12



PROJETO	**ROUTE ECU / ECTU**
STATUS	Em desenvolvimento
FOLHA / SUBSISTEMA	**Temperaturas e termopares**
AUTOR	Matheus Markies
ATUALIZAÇÃO	14/10/2026
REVISÃO	A definir
FORMATO	A3
CÓDIGO DO DESENHO	ECTU-SCH-12.
ARQUIVO	TemperatureSensors.SchDoc
FOLHA	12 / 12